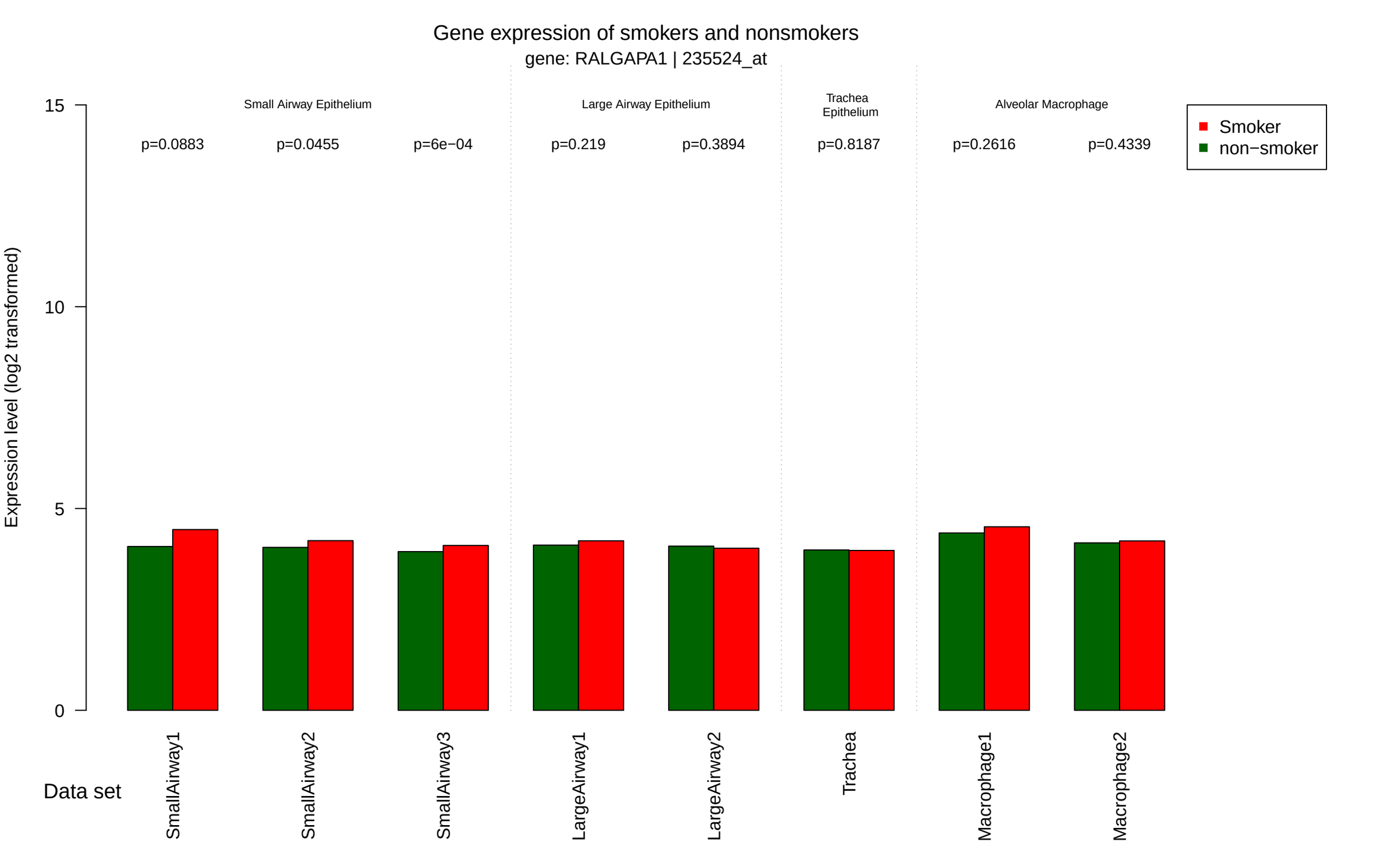
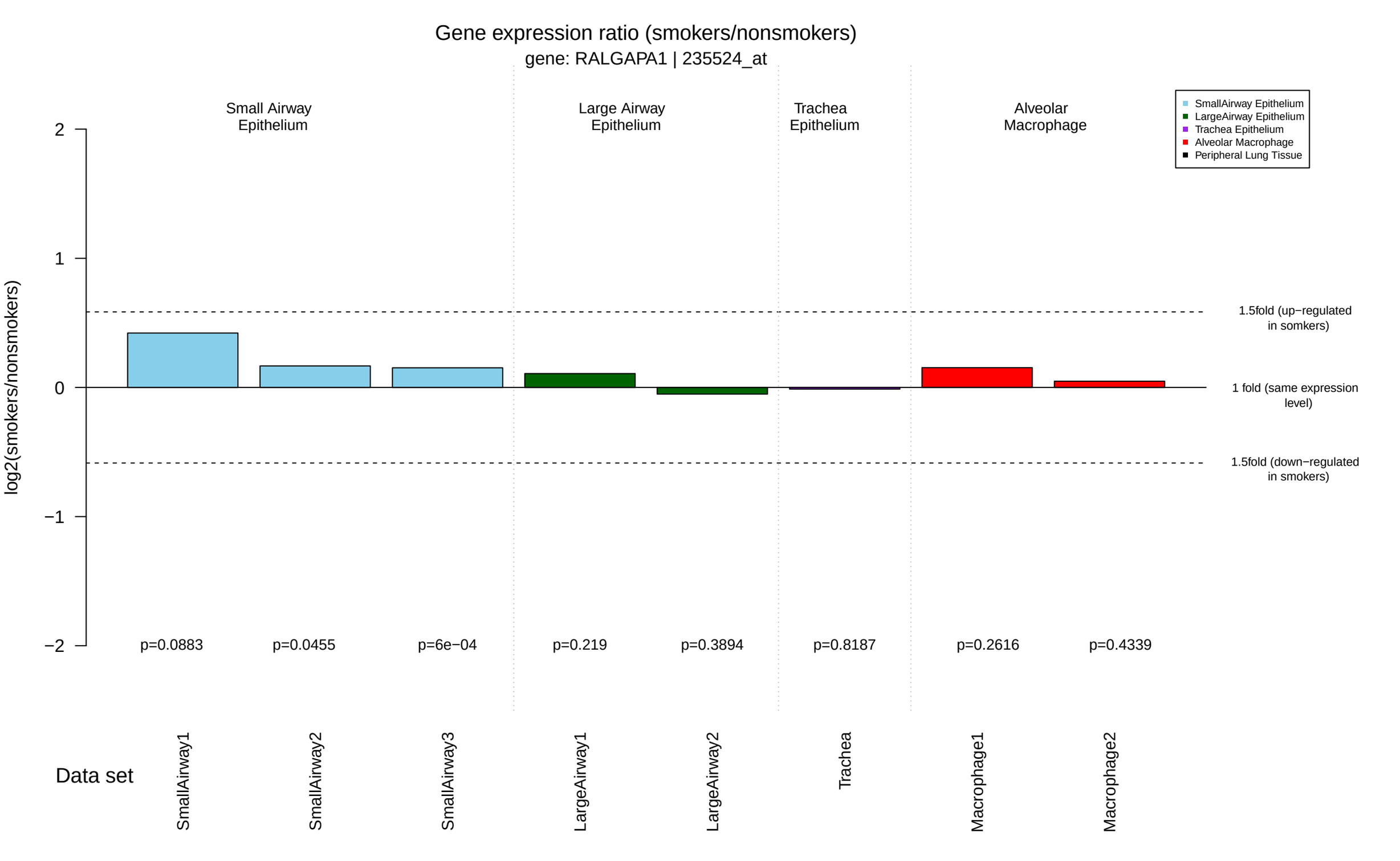
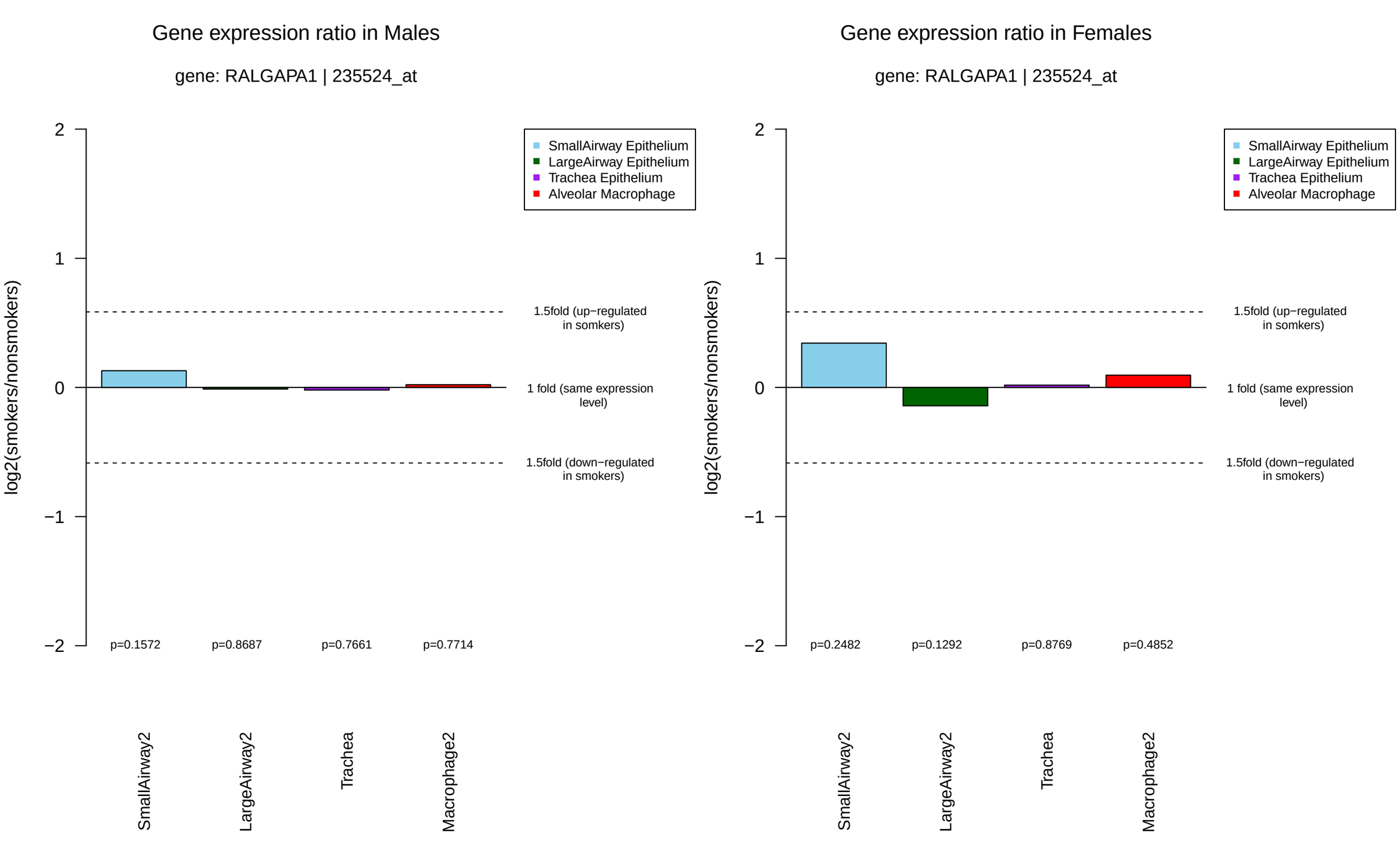
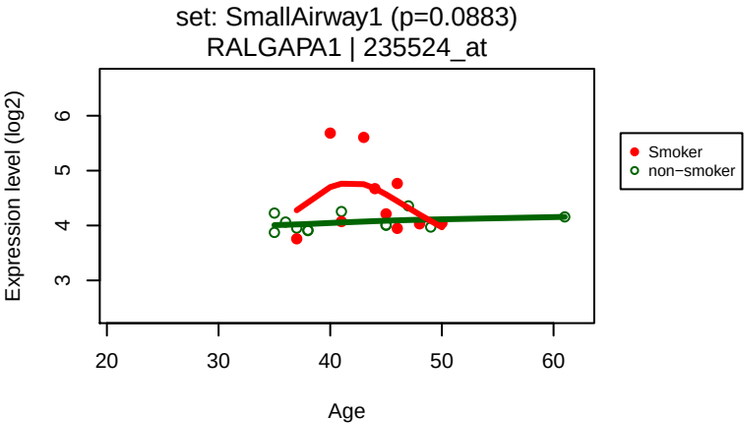
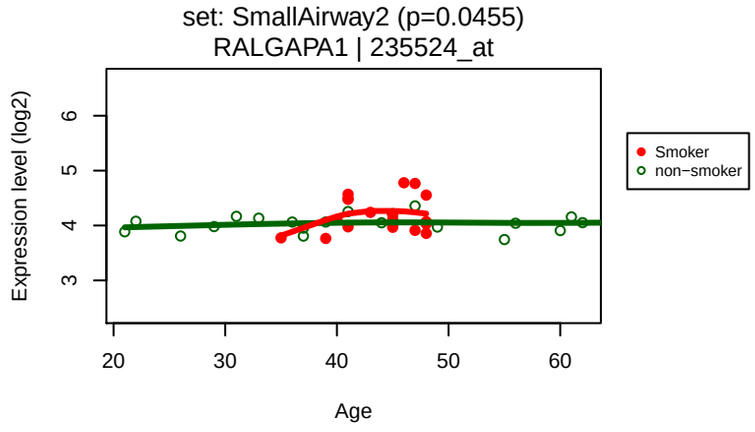
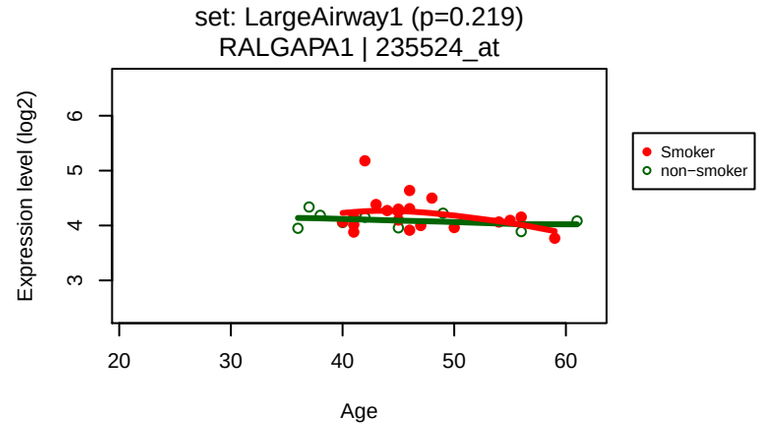

Gene expression of smokers and nonsmokers
gene: RALGAPA1 | 235524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0883
p=0.0455
p=6e−04
p=0.219
p=0.3894
p=0.8187
p=0.2616
p=0.4339
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

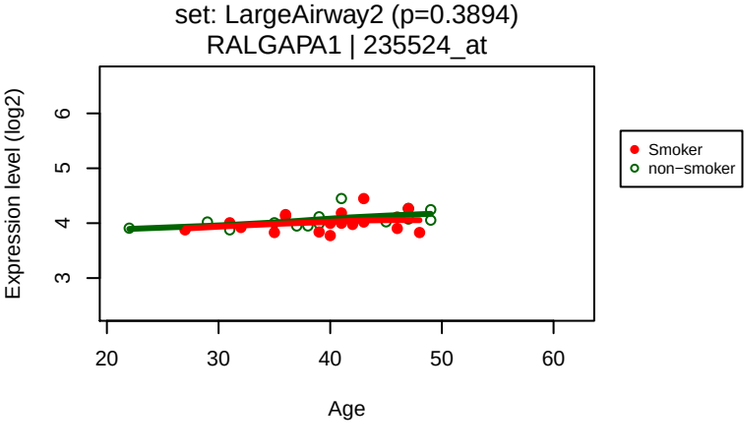

Gene expression ratio (smokers/nonsmokers)
gene: RALGAPA1 | 235524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0883
p=0.0455
p=6e−04
p=0.219
p=0.3894
p=0.8187
p=0.2616
p=0.4339
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

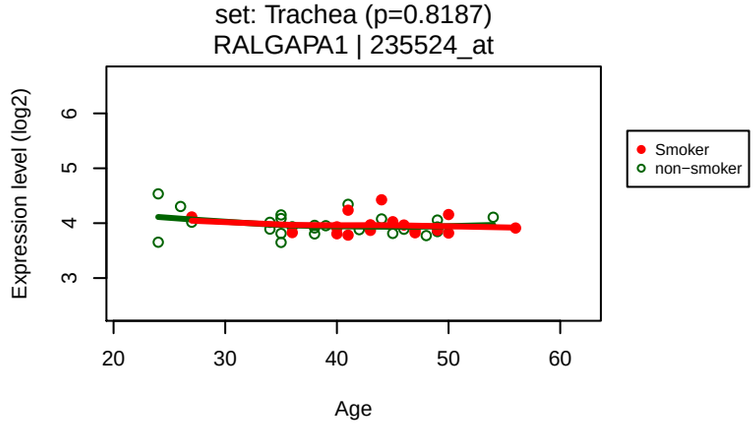

Gene expression ratio in Males
gene: RALGAPA1 | 235524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1572
p=0.8687
p=0.7661
p=0.7714
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALGAPA1 | 235524_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2482
p=0.1292
p=0.8769
p=0.4852
SmallAirway2
LargeAirway2
Trachea
Macrophage2

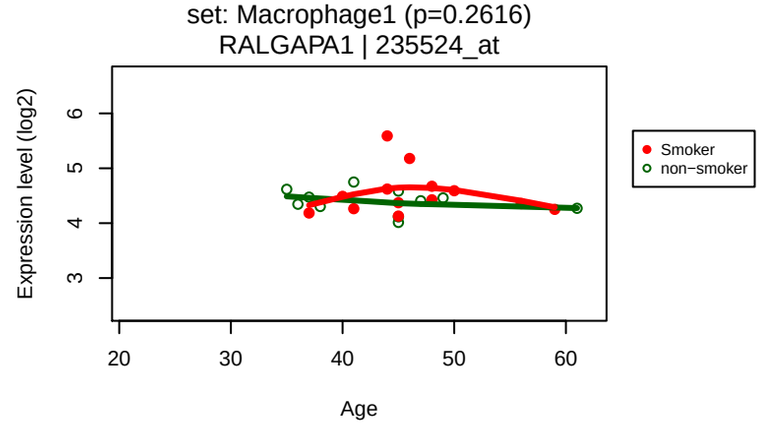

set: SmallAirway1 (p=0.0883)
RALGAPA1 | 235524_at
Expression level (log2)
Age
Smoker
non−smoker

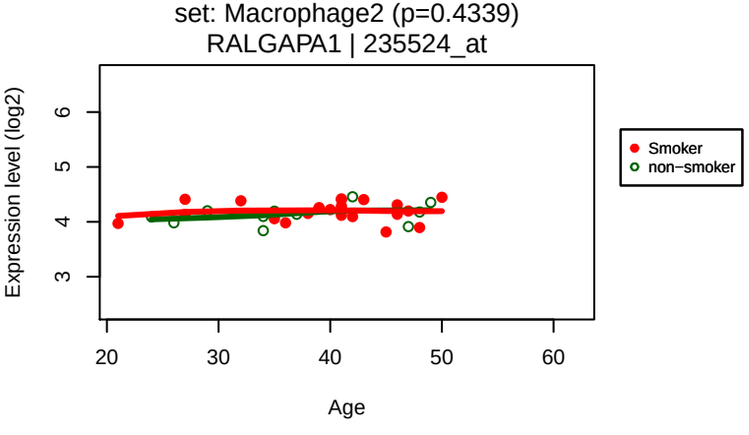

set: SmallAirway2 (p=0.0455)
RALGAPA1 | 235524_at
Expression level (log2)
Age
Smoker
non−smoker

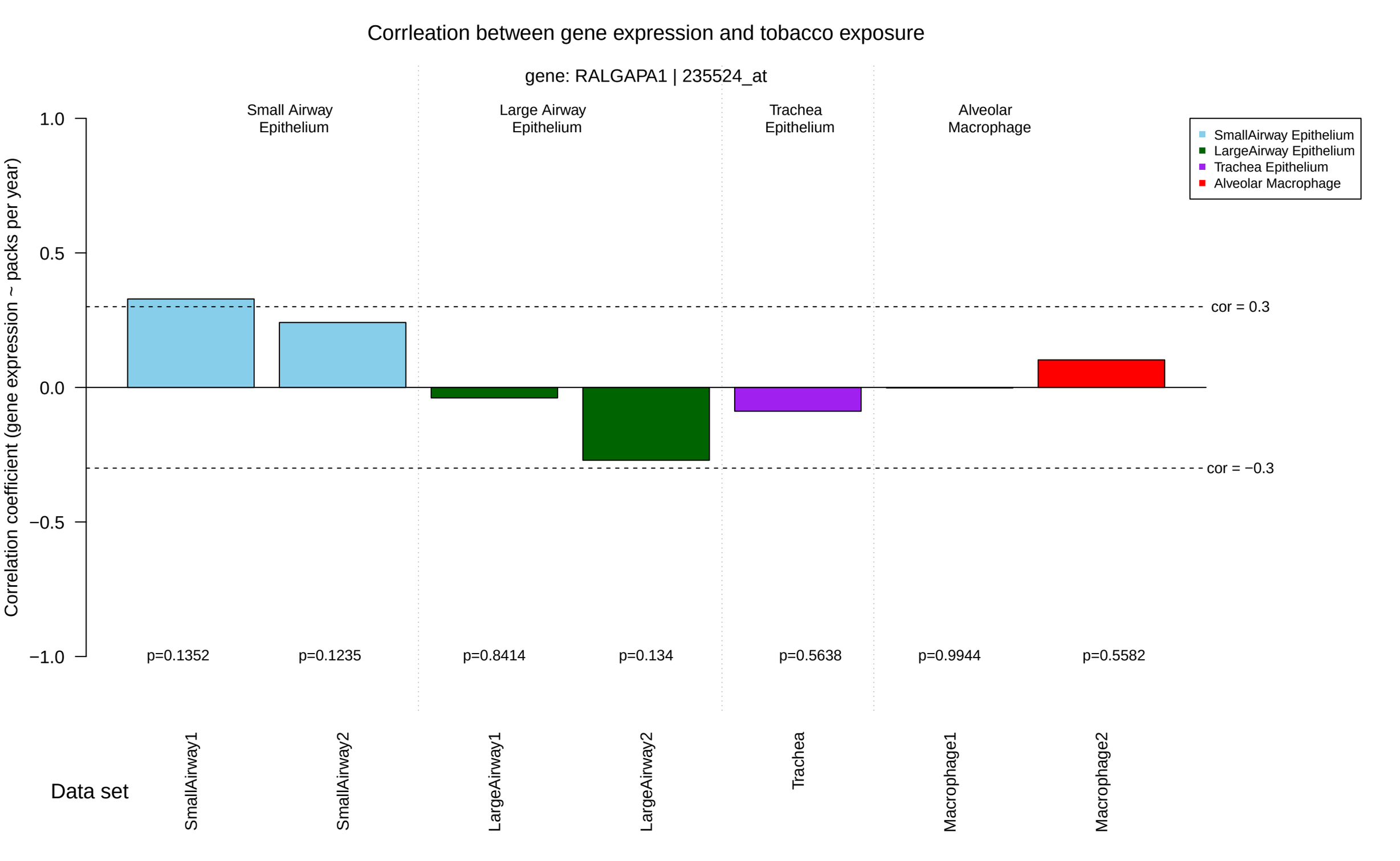

set: LargeAirway1 (p=0.219)
RALGAPA1 | 235524_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3894)
RALGAPA1 | 235524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8187)
RALGAPA1 | 235524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2616)
RALGAPA1 | 235524_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4339)
RALGAPA1 | 235524_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALGAPA1 | 235524_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1352
p=0.1235
p=0.8414
p=0.134
p=0.5638
p=0.9944
p=0.5582
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2